

**STUDYHUB**  
**CLASS 10 — MP BOARD**  
**SUBJECT: SCIENCE**  
**SAMPLE QUESTION PAPER — SET 1**

|                 |                   |
|-----------------|-------------------|
| Time: 3 Hours   | Maximum Marks: 75 |
| Medium: English | Set: 1            |

Note: This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

**GENERAL INSTRUCTIONS**

1. All questions are compulsory.
2. Questions 1 to 5 are objective-type questions. Each question carries 6 marks.
3. Internal choices are given from Questions 6 to 23.
4. Answer Questions 6 to 17 in about 30 words each.
5. Answer Questions 18 to 20 in about 75 words each.
6. Answer Questions 21 to 23 in about 120 words each.
7. Draw neat and labelled diagrams wherever required.
8. Balance chemical equations wherever required.

**PART — A**

**OBJECTIVE TYPE QUESTIONS**

Q.1 Choose the correct alternative. [6 Marks]

(1) Which type of reaction involves exchange of ions between two compounds?

- A. Combination reaction
- B. Displacement reaction
- C. Double displacement reaction
- D. Decomposition reaction

(2) A solution has a pH of 3. The solution is—

- A. Strongly basic
- B. Acidic
- C. Neutral
- D. Slightly basic

(3) Which of the following is a good conductor of electricity?

- A. Sulphur
- B. Graphite
- C. Phosphorus
- D. Rubber

(4) Which tissue mainly transports food in plants?

- A. Xylem
- B. Phloem
- C. Meristem
- D. Epidermis

(5) A lens has a power of +2 D. It is a—

- A. Concave lens
- B. Convex lens
- C. Plane mirror
- D. Concave mirror

(6) Which device demonstrates the magnetic effect of electric current?

- A. Electric bell
- B. Thermometer
- C. Barometer
- D. Hydrometer

Q.2 Fill in the blanks. [6 Marks]

- (1) The substance formed when magnesium ribbon burns is \_\_\_\_\_.
- (2) An acid generally reacts with a base to form salt and \_\_\_\_\_.
- (3) The valency of carbon is \_\_\_\_\_.
- (4) The main pigment required for photosynthesis in plants is \_\_\_\_\_.
- (5) Light-sensitive cells in the human eye are present on the \_\_\_\_\_.
- (6) The commercial unit of electrical energy is \_\_\_\_\_.

Q.3 Write True or False. [6 Marks]

- (1) Addition of oxygen to a substance can be called oxidation.
- (2) All metals are gases at room temperature.
- (3) Ethane is a saturated hydrocarbon.
- (4) The brain is always directly involved in a reflex action.
- (5) A convex mirror always forms a virtual image.
- (6) An electric motor converts electrical energy into mechanical energy.

Q.4 Match the following. [6 Marks]

| Column A         | Column B                                  |
|------------------|-------------------------------------------|
| (i) Baking soda  | (a) Myopia                                |
| (ii) Myopia      | (b) $\text{NaHCO}_3$                      |
| (iii) Insulin    | (c) Pancreas                              |
| (iv) Pollination | (d) Transfer of pollen grains             |
| (v) Ohm's law    | (e) $V = IR$                              |
| (vi) Ozone       | (f) Protection from ultraviolet radiation |

Q.5 Answer in one word or one sentence. [6 Marks]

- (1) Write the term used for the process in which a substance combines with oxygen during burning.
- (2) Name the tissue that transports water in plants.
- (3) What is the female gamete in humans called?
- (4) What is the SI unit of electrical resistance?
- (5) Name the defect in which distant objects are not seen clearly.
- (6) Give one example of a producer in a food chain.

## PART — B

### SHORT ANSWER QUESTIONS

Q.6 Write any two factors affecting the rate of a chemical reaction. [2 Marks]

**OR**

What is a catalyst? Give one example. [2 Marks]

Q.7 Write any two differences between acids and bases. [2 Marks]

**OR**

Write any two importance of pH in daily life. [2 Marks]

Q.8 Write any two chemical properties of metals. [2 Marks]

**OR**

Write any two properties of ionic compounds. [2 Marks]

Q.9 What is meant by a homologous series? [2 Marks]

**OR**

Why does carbon show the property of catenation? [2 Marks]

Q.10 Write any two functions of the small intestine in human digestion. [2 Marks]

**OR**

Write any two functions of blood. [2 Marks]

Q.11 What is a reflex action? Give one example. [2 Marks]

**OR**

What is phototropism? Explain briefly. [2 Marks]

Q.12 Write any two methods of asexual reproduction with examples. [2 Marks]

**OR**

Where does fertilisation normally occur in humans? [2 Marks]

Q.13 What are dominant and recessive traits? [2 Marks]

**OR**

Write one reason for the occurrence of variation in sexual reproduction. [2 Marks]

Q.14 What is the relation between the focal length and radius of curvature of a mirror? [2 Marks]

**OR**

Write any two examples of refraction from daily life. [2 Marks]

Q.15 What is the accommodation power of the human eye? [2 Marks]

**OR**

Which type of lens is used to correct myopia? Give the reason. [2 Marks]

Q.16 A  $6\ \Omega$  resistor is connected across a potential difference of 12 V. Find the current flowing through it. [2 Marks]

**OR**

Differentiate between electrical power and electrical energy. [2 Marks]

Q.17 Describe one experiment showing that electric current produces a magnetic field. [2 Marks]

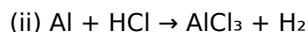
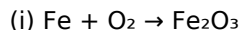
**OR**

On what energy conversion does an electric generator work. [2 Marks]

## **PART — C**

### **SHORT ANSWER QUESTIONS**

Q.18 Balance the following chemical equations:



**OR**

Differentiate between oxidation and reduction and give one example of each. [3 Marks]

Q.19 Briefly explain the process of respiration in humans. How does exchange of gases take place in the lungs? [3 Marks]

**OR**

Briefly explain the process of urine formation with the help of a nephron. [3 Marks]

Q.20 Why do stars appear to twinkle? Explain on the basis of atmospheric refraction. [3 Marks]

**OR**

Why does the sky generally appear blue? Explain on the basis of scattering of light. [3 Marks]

## **PART — D**

### **LONG ANSWER QUESTIONS**

Q.21 A student connects two resistors of  $4\ \Omega$  and  $6\ \Omega$  in series with a 10 V battery.

(i) Find the total resistance.

(ii) Find the current flowing through the circuit.

(iii) Find the potential difference across the  $6\ \Omega$  resistor. [4 Marks]

**OR**

Write the formula used to calculate electrical energy. A 100 W bulb is used for 5 hours per day for 30 days. Calculate the electrical energy consumed in kWh. [4 Marks]

Q.22 Draw a neat and labelled diagram of the female reproductive system in humans and briefly write the functions of its major parts. [4 Marks]

**OR**

Explain the process of sexual reproduction in humans from fertilisation to the development of the embryo in proper sequence. [4 Marks]

Q.23 Answer any one of the following: [4 Marks]

(i) Using a ray diagram, explain the image formed by a concave mirror when the object is placed between the focus and the centre of curvature. Also state the position, nature and size of the image.

**OR**

(ii) An object is placed 20 cm in front of a convex lens having a focal length of 10 cm. Using the lens formula, find the position of the image and state whether it is real or virtual. Draw the required ray diagram.

# ANSWER KEY

## Objective Answers

Q.1: (i) C (ii) B (iii) B (iv) B (v) B (vi) A

Q.2: (i) Magnesium oxide / MgO (ii) Water (iii) 4 (iv) Chlorophyll (v) Retina (vi) kWh

Q.3: (i) True (ii) False (iii) True (iv) False (v) True (vi) True

Q.4: (i)-(b), (ii)-(a), (iii)-(c), (iv)-(d), (v)-(e), (vi)-(f)

Q.5: (i) Oxidation (ii) Xylem (iii) Ovum (iv) Ohm ( $\Omega$ ) (v) Myopia (vi) Green plant / Grass

## Numerical Answers

Q.16:  $I = V/R = 12/6 = 2 \text{ A}$ .

Q.21: Total resistance =  $10 \Omega$ ; current =  $1 \text{ A}$ ; potential difference across  $6 \Omega = 6 \text{ V}$ .

Q.21 (OR): Electrical energy consumed =  $15 \text{ kWh}$ .

Q.23 (ii): Image distance =  $20 \text{ cm}$ ; the image is real, inverted and of the same size.

## MARKS VERIFICATION

| Section          | Questions    | Marks |
|------------------|--------------|-------|
| Objective        | 1-5          | 30    |
| Short Answer     | 6-17         | 24    |
| 3-Mark Questions | 18-20        | 9     |
| Long Answer      | 21-23        | 12    |
| TOTAL            | 23 Questions | 75    |

Final check:  $30 + 24 + 9 + 12 = 75$  Marks.